FitBuddy

Sample Project Documentation

AI Fitness Plan Generator using Gemini models | Draft sample based on project brief

1. Project overview

FitBuddy is a web application concept that helps users create and refine a basic 7-day workout plan. It uses user-provided fitness goals and preferred intensity to personalize suggestions, with a short nutrition or recovery tip.

2. Problem and proposed solution

Problem: People may have a fitness goal but lack a clear starting routine. Generic plans can be difficult to follow or adjust as preferences change.

Solution: Provide a structured weekly plan, accept user feedback, and generate an updated plan while keeping the original version for reference.

3. Objectives

Make fitness planning easier to start; tailor workouts to submitted preferences; support revisions; save plan history; give coaches an authorized overview.

4. Main functional features

1. User form and input validation. 2. Personalized 7-day workout plan.
3. Daily activity and recovery details. 4. Goal-related nutrition/recovery tip.
5. Feedback-based plan update. 6. Save original and updated plan versions.
7. Readable results page. 8. Coach/admin user and plan overview.

5. Intended users

Primary: people seeking a basic fitness routine. Secondary: coaches or administrators reviewing user plans. The product is a planning aid, not medical advice.

FitBuddy

Design, Setup, and Quality

6. Technology and architecture

Frontend: HTML, CSS, and Jinja2 templates. Backend: Python and FastAPI.
AI: Gemini API for workout creation, plan revision, and short tips.
Data: SQLite with SQLAlchemy. Local server: Uvicorn.

Flow: browser form -> FastAPI -> AI service and SQLite -> rendered result page.

7. Data and privacy

Expected records include user details, goal, intensity, original plan, feedback, updated plan, and timestamps. Collect only what the app needs, protect API keys, and restrict the admin view to authorized users.

8. Local setup and use

```
python -m venv venv
venv\Scripts\activate
pip install -r requirements.txt
uvicorn app.main:app --reload      Use the import path that matches the app.
```

Configure GOOGLE_API_KEY outside source control. Open 127.0.0.1:8000 in a browser.

9. Testing plan

Check valid/invalid form inputs, plan and tip generation, database persistence, feedback updates, original plan retention, admin access, and error handling.

10. Limitations and future work

AI suggestions may be inaccurate and depend on external service availability.
Future work: authentication, stronger safety preferences, progress tracking, and deployment beyond a local development environment.